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An overview of the way organic farming is evolving in sericulture

Chakrapani* & Rashmi Mishra

University Department of Botany, Dr. Shyama Prasad Mukherjee University, Ranchi, Jharkhand, India

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Abstract- India is unique in that it is the only country that produces all four types of silk: mulberry (79.23%), eri (13.32%), tasar (6.8%), and muga (0.54%). The sericigenous tasar silkworm is a polyphagous species that consumes more than 24 different types of plants, including *Shorea robusta, Terminalia tomentosa,* and *Terminalia arjuna*. Eri silkworms feed on the leaves of several different plants, including Kassava, Kesseru, Castor, Tapioca, and Payam. The fragrant leaves of the som (*Persea bombycina*) plant are the main food source for muga silkworms, which are raised on trees. They can also consume mejankari (*L.citrata*), dighloti (*L. salicifolia*), and soalu (*Litsea monopetala*) in the absence of som. The leaves of the Morus indica mulberry plant are used to raise mulberry silk. The mulberry tree, a perennial with widely dispersed branches, is specifically grown to feed the silkworms. Superior organic manures are essential for the growth and quality of leaves. Aeration, optimal permeability, soil aggregation, rooting depth, and a decrease in bulk density, erosion, and soil crusting are all signs that organic fertilizers, such as FYM and biofertilizers, improve the physical properties of the soil and serve as soil conditioners, supplying plant nutrients. Over time, organic nutrient management techniques have changed, and different plant species now require different mixes of organic manures and composts, along with techniques like mulching and green manuring. Chicken dung is observed to perform better when it comes to boosting the amount of free amino acids, total soluble sugars, total reducing sugars, total soluble proteins, and chlorophyll content in leaves. Poultry manure, fish meal, pig manure, farmyard manure, and press mud are among the many organic manures that India has to offer as nitrogen sources.

Keywords: FYM, Green Manure, Biofertilizer, Poultry manures

INTRODUCTION

The process of raising silkworms to produce silk fibers is known as sericulture. Sericulture is the humane method used to raise silkworms for the production of silk. The Greek word "Sericos," which means silk, and the English word "culture," which means upbringing, are the sources of the phrase sericulture.[1]

India holds the unusual distinction of being the only nation that produces all four types of silk: muga (0.54%), tasar (6.8%), eri (13.32%), and mulberry (79.23%). In addition to being the world's biggest consumer of silk, India is the world's second-ranked silk producer. Mulberry sericulture is mainly practised in states such as Karnataka, Andhra Pradesh, Assam and Bodoland (Kokrajhar, Chirang, Baksa and Udalguri districts of Assam), West Bengal, Jharkhand and Tamil Nadu who are the major silk producing states in the country. North East has the unique distinction of being the only region producing four varieties of silk *viz.*, Mulberry, Oak Tasar, Muga and Eri. Overall NE region contributes 18% of India's total silk production.

The two primary proteins produced by the silkworm as the major building blocks of the cocoons of silk fiber

*Corresponding author :
Phone : 74886116807
E-mail : r.chakrapani2221@gmail.com

are sericin and fibroin. These unique structures are perfect for the metamorphosis of larvae into adults. The sericigenous tasar silkworm is a polyphagous species that feeds on over 24 different food plant species in addition to *Terminalia tomentosa, Terminalia arjuna, and Shorea robusta.*[2-4] It is important to maintain the silkworm host plant in order to produce high-quality cocoon. In order to obtain enough leaves, food plants have been evaluated for silkworm rearing based on their availability, accessibility, quantity, crop season, plant development, and compatibility. Furthermore, a crucial element that is influenced by soil fertilization, plant spacing, and variety is leaf quality.[5,6] The function that nutrition plays in sericulture is significant. The silkworm's health and growth, as well as the commercial qualities of the silk that is produced, are determined by the nutritional quality of the leaves that it feeds on. Therefore, mulberry leaves determine the quantity and quality of raw silk produced as well as the subsequent growth of the sericulture industry.

Variations in leaf moisture, nutrition, and biochemical components are reflected in cocoon characteristics, and leaf quality in mulberry types affects the degree and homogeneity of moulting.[6,7] In general, nutrient-rich foliage guarantees superior larval growth rates, higher weights of larvae, and a higher moulting ratio.[8] The primary source of protein for the manufacture of silk, rich protein leaves make silkworms quite heavy.[9]

Studies have shown that the suitability of host plants strongly influences the growth, development, and reproductive potential of *P. ricini.* Host plants with different foliar constituents and nutrient compositions significantly affect important economic parameters in eri silkworms.[10-15]

In addition, assessed 23 plants besides major hosts and identified the fountain tree (*Spathodea companulata* Pal.) and carrot (*Daucus carota* Linn.) as appropriate hosts. In earlier attempts, tapioca/cassava, *Manihot esculenta* Crantz (= *Manihot utilissima*), payam, *Evodia flaxinifolia* Hook, and Kessuru, *Heteopanax fragrans* (Roxb.) were suggested for rearing.[16,17] The plants with the highest volumetric qualities were castor, followed by tapioca, ailanthus (*Ailanthus excelsa* Roxb.), plumeria/frangipani (*Plumeria acutifolia* Poir), and papaya (*Carica papaya* Linn.).[16,18] In conclusion, castor is the best plant to raise silkworms because its leaves have higher levels of nitrogen and amino acids than those of other host plants.[19] Additionally, the silkworm life cycle was completed on tapioca, castor, *P rubra,* and *A. excelsa* in 46, 63, 62, and 72 days, respectively.[20]

Host plants have an impact on the feeding and moulting habits of muga silkworms, which in turn affects the features of the cocoon.[21] According to field research, the leaf quality of different genotypes/varieties of som, soalu, and other natural host trees can influence the growth and development of muga silkworms.[22-24] There have been reports of improved grainage metrics, increased cocoon weight, and higher shell % in soalu-fed larvae and som-fed larvae, respectively.

The silkworm has a predilection for food because of the connection between it and the feeding plant. Primary food/host plants are those that are typically preferred by silkworms. Secondary food/host plants are other types of plants that silkworms can survive but often do not prefer.[25] In tasar silkworms, feeding on nutrient-rich leaves can improve larval growth, ERR, and cocoon yield, which in turn can improve other raw silk commercial features.[26] The food plant's components have a significant impact on the insect's ability to produce silk. These factors include survival behaviour, the rate at which food is consumed, digestion, and assimilation, all of which have a direct impact on the silkworm's ability to grow and develop.[27-31]

Organic Farming:

An essential natural resource for the production of crops is soil. The ability of a soil to guarantee the availability of nutrients necessary for plant growth and development is referred to as soil fertility. A variety of factors influence crop yield magnitude, fertility being just one of them. One of the main obstacles to the best possible crop development and output is low fertility. Fertilizers can be used to control soil fertility, but in order to choose the right type and amount of fertilizer to use, farmers need to understand the type and severity of the nutrient problem.

Organic farming has been popular as a solution to these issues since it uses natural resources like organic fertilizers instead of toxic substances like pesticides, herbicides, and fertilizers.[32] Pesticide residues in food are a result of the modern inorganic farming system's use of synthetic fertilizers, which also causes rapid destruction of soil structure, a decreased response to continuous fertilizer use, the loss of beneficial microorganisms and subsequent microbial nutrient circulation systems, especially toxic pesticides. Vermicomposting is a quick and

easy way to recycle kitchen scraps, city trash, and agricultural waste. It also uses earthworm action to bioconvert organic waste materials into nutrient-rich compost.[33] Result of these, the effects of vermicompost on plant growth have been well documented, yet it is mostly employed as a source of "Nitrogen" and "Phosphorus," which are important nutrients that are part of several essential plant structural components and act as catalysts in a variety of important plant biochemical reactions. The term "farmyard manure" describes the decomposing combination of farm animal feces and urine, litter, and leftover materials from roughages or cow feed. Farmyard manure that has decomposed well typically comprises 0.5% N, 0.2% P_2O_5, and 0.5% K_2O. Farmyard manure (FYM) enhances the physical, chemical, and microbiological aspects of the soil in addition to providing growth-regulating chemicals.[34-37] In crop-livestock farming systems, farmyard manure is easily accessible. By giving plants nutrients, especially micronutrients, and enhancing the physical, chemical, and biological characteristics of the soil, FYM treatment has been shown to promote crop development.[38] Increase in rice root length and root volume with FYM application, indicating that the better root development would allow the plant to exploit more water under water stress conditions.[38] FYM improves the soil structure, which in turn creates a better environment for root development.

Crop production is increased by using organic and biofertilizer because they enhance the humic components in the soil.[39] The application of Phosphate Solubilizing Bacteria (PSB) has been shown to positively impact plant height.[40] Bio-fertilizers increase soil microbial activity and release micronutrients that are not available for plant growth.[41] These fertilizers are components of primary and secondary nutrients that plants get during the mineralization process in an accessible form. Additionally, organic manures contain organic matter that helps the soil's aeration, structure, and ability to store water. By giving them energy, it also makes a lot of soil microbes more active. They improve the solubility of soil minerals and mineral nutrients in the soil, as well as the soil's capacity for buffering and exchange. Increased root growth is the outcome of using organic fertilizer, which also enhances cation exchange capacity and encourages soil aggregates. In addition to giving plants the necessary biological processes, organic fertilizers also help to control the proliferation of plant pests.

Relation among sericulture and organic farming:

The mulberry plant is a perennial that grows quickly, has deep roots, and produces a lot of biomass in its leaves. It serves as *Bombyx mori* L., the silkworm's primary dietary source. Increasing the amount of raw silk produced is mostly dependent on timely provision of high-quality mulberry leaves to silkworms. Consequently, it is evident that mulberry leaves are the primary source of sustenance for silkworms and play a significant role in the creation of cocoons. Studies have indicated that by implementing enhanced packaging and irrigation techniques, mulberry yields can be raised.[42] Mulberry tree cultivation packages and procedures have been provided for temperate regions.[43,44] Studies on the effects of nitrogen on mulberry development and yield have been conducted.[45] The study examined how mulberry saplings responded to the application of biofertilizers.[46] In Kashmir, different mulberry genotypes under rainfed conditions were evaluated.[47,48] Methods for soil sampling, testing, and management of various nutrient deficiencies in mulberry plants grown in temperate climates have been provided.[49,50]

Many elements, including biotic and abiotic components, agronomic techniques, and variation, affect the quality of mulberry leaves. Chemical fertilizers are typically employed in mulberry production to preserve and improve the quality of the leaves and growth. On the other hand, prolonged and frequent application of chemical fertilizers degrades the properties of the surface soil and impacts the availability and uptake of nutrients by plants.[51] Nitrogen fixers can improve and enrich the nutrient value of any compost, making it easier to substitute or supplement conventional fertilizers. Optimizing the quality of leaf yield is one of the key elements in a successful and high-quality cocoon production process. Due to their high nutrient content, concentrated organic manures have the potential to completely replace inorganic fertilizers in terms of nutrients. Applying organic manure enhances the physical, chemical, and biological characteristics of the soil, which directly affects nutrient conservation, root development, and moisture retention, among other things. Two percent of the nitrogen in chicken dung is in the form of uric acid, which quickly transforms into ammonia, making it easily available for plant uptake.[52] Poultry manure, fish meal, pig manure, farmyard manure, and press mud

are among the many organic manures that India has to offer as nitrogen sources. These are produced by 150 million chickens, 8.6 million pigs, 182 million, and 700 sugar mills, which will have doubled in number by the year 2000. Poultry dung is a rich source of nutrients in addition to acting as a substance for conserving soil.[53] The concentration of soluble phosphorus in the soil rose when poultry manure was applied at higher rates. Due to its rapid fermentation and simultaneous excretion of liquid and solid waste without urine loss, poultry dung is an abundant source of nutrients.

An effort was conducted to determine how manures affected the biochemical components of mulberries. It's possible that applying the poultry manure enhanced the release of macro and micronutrients, which in turn enhanced the production of dry matter, plant height, and nutrient uptake, ultimately increasing the yield of mulberry leaves. The mulberry leaf's ability to absorb nutrients has risen due to the application of poultry manure. Poultry manure's higher total and accessible nitrogen content may have contributed to the greater absorption. Higher nitrogen intake was caused by increased nitrogen content and dry matter generation. Uric acid, which quickly transforms into ammoniacal form for efficient plant uptake, makes up 60% of the nitrogen in poultry manure.[52,54] Enhanced physico-chemical characteristics of the soil and a consistent, sufficient supply of nitrogen could have contributed to the higher uptake of nutrients by poultry manure in various crops. The presence of poultry manure boosted the chemicals that promote growth, perhaps leading to improved plant development and increased nutrient uptake. The high level of phosphorus availability in the chicken manure was a significant factor that helped or affected the crop to produce superior growth and yield components.[55] The complete set of important plant nutrients-nitrogen, phosphorus, potassium, calcium, magnesium, sulphur, boron, zinc, copper, manganese iron, etc that are required to boost leaf yield and quality are clearly present in poultry manure.[56] Poultry excretes both liquid and solid parts; the concentrated nitrogen and phosphorus found in poultry manure is one of the main nutrients it contains. The outstanding fertilizing properties have been well-documented.[57,58] Mulberry leaf moisture content is a hereditary trait that is influenced by root growth and the amount of moisture in the accessible soil. The silkworm's digestibility is determined by the leaf's moisture content.[59] Mulberry leaves have a moisture content that ranges from 64 to 83%,[60] with 70% or more being regarded ideal[61]. The thickest cocoon was generated by silkworms that were fed on leaves with a higher moisture content (75%).[62] The moisture content of the applied organic manure and control plant leaves ranges from 71.25% to 76.76%. From the perspective of silkworm nutrition, mulberry leaves with higher moisture, protein, sugar, and carbohydrate content and lower mineral and crude fiber content are the best.[63] Numerous studies have demonstrated the value of chemical fertilizers and farmyard manure in the production of mulberries.[64-66] The recommended dosage of fertilizer and manure affects the quantity and quality of mulberry leaves produced, however farmers do not always apply these practices, which has a negative impact on productivity and output.[67] Recommended management approaches in agricultural fields should be adapted to enhance soil fertility in temperate climates.[68]

Indiscriminate use of chemical fertilizers negatively impacts the microflora and chemical characteristics of the soil related to mulberries; therefore, appropriate water and nutrient management is necessary to attain maximum production.[69] In respect to different inputs utilized by farmers at different intervals, researchers had demonstrated the significance of integrated nutrient management (INM) in soil fertility management at different farming stages.[70] There is a need to evaluate the availability of micronutrients and their relationship to the chemical characteristics of soil.[71-73] The importance of nutrient availability and interaction has also been highlighted. There have also been reports of variable silkworm cocoon crop performances linked to leaf nutrient deficiencies.[65,74] It has been suggested that appropriate fertilizers should be applied based on soil fertility evaluation to improve the productivity of mulberry gardens.[75] Research has been done on soil fertility status and the relationship between several fertility metrics.[76,77] The industry has recently adopted the use of nano fertilizers, which release nutrients gradually. Additionally, reports on the relationship between the metrics of leaf quality and soil fertility are available.[75,78] Two studies have demonstrated the significance of organic fertilizer input in maintaining soil health.[79,80] There are ways to improve Kashmir's mulberry wealth and enhance the rooting potential of temperate mulberry cultivars by FYM.[48] Previous studies have demonstrated that soil microflora play a significant

role in activities that promote plant growth.[81-83] Research was conducted on the effectiveness of utilizing Rhizobium and AM fungi in temperate environments.[84] It follows that there are enticing opportunities for the long-term storage of organic carbon in soil in temperate ecosystems.

Organic nutrients and their supply in sericulture: Organic nutrients are essential to the growth and development of mulberry plants because they assist the plants withstand both biotic and abiotic challenges, such as pest and disease attacks and high temperatures.[85] Vermicompost is a great source of plant nutrients, hormones, and enzymes for plant growth. It also improves mulberry quality and yield.[85] The number and length of leaves and shoots on mulberry plants were enhanced by the vermicompost treatment. Additionally, superior larval and cocoon characteristics, decreased larval mortality, and enhanced silk yield were seen in silkworms fed mulberry leaves treated with vermicompost.[86] Vermicompost treatment of mulberry plants has been shown to boost larvae weight, reduce illness incidence, and increase rearing rate.[87] Similar results were observed, showing that vermicompost treatment enhanced plant growth and improved economic returns.[88]

Vermiwash is the liquid vermicompost extract made on an earthworm medium with a high population density. In addition to a large population of decomposing bacteria, it is rich in vitamins, hormones, and antimicrobial peptides.[89] Vermiwash has been reported to be very effective supplement for *Bombyx mori*. The studies conducted to evaluate the efficiency of vermiwash on mulberry plants and thereby the silkworm reports the positive results. Vermiwash when sprayed on mulberry plants showed significant increase in growth of plant, leaf wield, which accounted to increase in larval weight, silk gland, cocoon and pupal weight as well as weight of shell of silkworm.[90,91] Vermiwash has been reported to enhance not only the quantity of buds and leaves as well as their weight, but also the biochemical constituents of the leaves, such as protein and carbs.[92] Vermiwash can be utilized as a feed additive for silkworms in sericulture to achieve good performance in economic attributes.[93] The weight increase of the larvae, the volume of the silk gland, the pupae, the shell, the pupal weight, and the production of the silk have all been documented by the authors. Both the growth and nutrient contents of mulberry leaves were shown to be greatly increased by vermicompost and vermiwash.[94]

Phosphorus, nitrogen, calcium, magnesium, potassium, cellulose, nemicellulose, lignin, and micronutrients like as zinc, copper, iron, manganese, etc. are abundant in poultry manure droppings. As a result, the manure produced when chicken faeces breaks down is an excellent source of organic nutrients.[85] Due to the increased mineral content in the leaves of mulberry bushes grown with chicken manure, adding poultry manure to the soil can be advantageous when raising silkworms.[95]

Dalweed is easily obtained organic manure that can be found in huge quantities in Dal Lake, Kashmir. This has a high nitrogen, phosphorus, and potassium content and may provide plants with nourishment. Dalweed is applied as a manure to promote seed germination, soil fertility, and plant productivity.[96] Among all the manures evaluated, dallweed proved to be the most effective for mulberry growth and seed germination.[97] Additionally, using dallweed can relieve the strain on other manures, give stakeholders a variety of options, and be a cost-effective and environmentally responsible way to dispose of waste.

Natural substances that contain living microorganisms are known as biofertilizers. They serve as an alternative to synthetic fertilizers, which encourage plant development by making nutrients more accessible or available. Using biofertilizers is a great method to advance sericulture. The use of biofertilizers increased filament size, weight, and length, and also enhanced the weight of juvenile cocoon shells and silk glands.[98] The optimal pupation ratio and cocoon formation percentage values were obtained with the use of biofertilizers.

Oil cake is the coarse leftover soil left over after oilseeds have been extracted of their oil. Rich in protein and minerals, oil cake has been shown to increase the nutritional content of mulberry leaves while lowering the spread of disease and pests. It was discovered that combining oilseed cakes of groundnut, castor, neem, and sesame with extracts of pugamia and turmeric effectively increased the survivability, growth, and yield of mulberry plants. Moreover, root rot diseases may be managed with neem oil cake.[99] Mixed oil cake, press mud, and potassium humate greatly enhanced the nutritional content and yield of mulberry leaves.[100]

Farm yard manure is an organic manure that is produced by breaking down animal excrement, such as cow or buffalo dung, and extra feed that is kept in pits.[101] It has an abundance of nutrients. In order to determine

how well press mud and farm yard manure affected the quality, development, and production of mulberry plants as well as how they affected silkworms.[102] The authors noted an increase in leaf production as well as an improvement in the leaves' nutritional quality, including moisture content, total soluble protein, sugar, nitrogen, and phosphorus content, and 216 chlorophyll concentration. Furthermore, the weight, yield, and shell percentage of the cocoons all significantly improved. Additionally, a high cost-benefit ratio was noted.

Comparing double hybrid silkworms to cross-breed silkworms, the study found that the application of liquid organic manures to the soil had a substantial impact on the economic features of the former. To understand the impact on cocoon quality, liquid organic manures must be applied consistently over an extended period of time. The use of chemical fertilizers to soil and its effect on silkworms reduces their capacity to generate high-quality cocoons. Thus, using liquid organic manures over an extended period of time can undoubtedly pave the road for sustainable and organic sericulture.[103]

Mulching helps mulberry (Variety SI) grow faster and provide higher-quality leaves by creating a favourable environment for the efficient use of organic manures and biofertilizers. Reducing the dosage of inorganic fertilizers and adding organic and biodegradable amendments to mulched plots were crucial in enhancing the physical, chemical, and biological properties of the soil and maximizing mulberry (*Morus alba* L.) yield potential. Therefore, for sustainable mulberry farming, the use of organic mulches in water-stressed conditions combined with the methodical use of organic fertilizers, including biofertilizers, may be advised.

Singhvi found that vermicompost has been beneficial not only for mulberry silk but also for Tasar silk.[104,105] Vermicompost has been shown to have positive effects on the quality and yield of *Terminalia tomentosa*, the Tasar silkworm's host crop, as well as on the production and conformance of cocoon crops. Furthermore, toward the end of the season, the plants that received vermicompost treatment still had leaves on them.

The ongoing use of tasar food plants for the rearing of tasar silkworms and soil erosion from careless fields have an impact on the soil's nutrient status, which in turn affects the production of leaves and cocoons. Thus, sunnhemp, dhaincha, and other crops are sown on tasar plantations, and green manuring is carried out by letting the crops mature to a certain length and incorporating them into the soil. It is anticipated that these green manure crops will lessen soil erosion while simultaneously enhancing soil structure, fertility, and water-holding capacity.Green mulching using leguminous crops such as sunhemp (*Crotalaria juncea*), daincha (*Sesbania aculeata*), green gram (*Vigna radiatus*), and black gram (*Vigna mungo*) was evaluated during the rainy season in a *Terminalia tomentosa* plantation.[106] They discovered that the mulching improved the soil's capacity to hold water, retain moisture, conduct electricity, and contain organic carbon, nitrogen, and potassium, while reducing bulk density and pore space. Sunhemp mulching improved the tasar field soil characteristics, *T. tomentosa* leaf yield, cocoon yield, and silk productivity of the tasar silk insect.[107] This made tasar culture more sustainable.

The practice of vermicomposting has spread over the globe however it is most well-liked in India.[104] This is the method by which organic materials are transformed by earthworms into vermicompost, a humus-like substance that is rich in nutrients and humus because it is the earthworms' excrement.[86,104] According to a report, applying freshly prepared vermicompost at a rate of 2.0 kg per plant to asan plants improved their leaf yield by 8.95% compared to the control group.[104] Additionally, the leaves of treated plants were available for a longer period of time towards the end of the season, or during the third crop, than those of the control group. Additionally, growing tasar silkworms on plants fed vermicompost increased the cocoon output by 9.03%.

An integrated pest management (IPM) package aimed at increasing crop output effectively reduces crop losses caused by gall insects in tasar host plants by combining cultural, mechanical, and chemical control practices.[108] Integrated leaf gall control in tasar decreases infestation and promotes the availability of high-quality leaves. Plant pruning can be postponed, diseased leaves can be removed and burned, and azadirachtin-based pesticides can be sprayed or caked. Additionally, herbicides based on neem were developed to control caterpillars that consume bark.

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